INCH-POUND

MIL-C-24643/9C
14 March 1994
SUPERSEDING
MIL-C-24643/9B
3 October 1986

MILITARY SPECIFICATION SHEET

CORD, ELECTRICAL, TYPE LSMRI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor - class K stranding, uncoated (see table I for size).
Second - Ethylene propylene rubber or cross-linked polyethylene insulation. Standard identification code shall be applied by method 3.
Third - The specified number of conductors cabled with a lay not greater than that specified (see table I)

Table I. Details

Military part no. M24643/9	Type and size	Conductors		Diameter over insulation		Lay of conductors maximum (inch)	Conductor resistance 1000 feet (max ohms)
		No.	Size AWG	Minimum (inch)	Maximum (inch)		
-01UN	LSMRI-D-1	2	20(class K)	0.068	0.075	2	11.05
-02UN	LSMRI-D-2-1/2	2	18(class K)	.092	.102	3	4.36
-03UN	LSMRI-T-2-1/2	3	16(class K)	.092	.102	3	4.36

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - conductor to conductor, volts, root mean square, minimum	
AWG 20 Class K	1000
AWG 16 Class K	1500
Insulation resistance - megohms/1000 feet, minimum	100
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Heat distortion - percent of unaged, maximum	30
Group D:	
* Flame propagation (cable)	Not Applicable

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Acid gas equivalent (insulation) - percent, maximum.	18
Smoke index, (insulation) maximum.....	45
Toxicity index, (insulation) maximum.....	1.5

UNIT ORDERING LENGTH:

Type	Size	Feet (nominal)
LSMRI-D-	1	600
LSMRI-D-	2-1/2	600
LSMRI-T-	2-1/2	500

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - AV, CR, ER, ME, AR, AL
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